RBPA - D2: Research-Based Preparatory Activity

Designing an ETL System Before Implementation

Serbisure

System Background

SerbiSure is a web-based recruitment platform connecting homeowners in Cagayan de Oro City with verified domestic workers (Kasambahay). Users include homeowners (employers), workers (job seekers), a superadmin (verification manager), and barangay stakeholders (monitoring). Primary purpose: Safe hiring via document checks (NBI/Police clearances), NLP for skill matching, sentiment-based reputation scores, and labor trend reports.

It addresses informal hiring issues (e.g., via Facebook) in a growing city, enabling organized matches, verified documents, and community insights.

Data Landscape

Currently, recruitment data exists in multiple fragmented sources. Workers provide resumes, personal info, and NBI clearances through paper documents or Facebook chats. Employers share job posts and requirements via Facebook groups or Messenger. Barangay offices keep worker registry forms on paper. The superadmin handles manual checks using spreadsheets or email. The data is unstructured like chat messages, semi-structured like resumes and job descriptions, and structured like barangay lists.

This setup means data is scattered and hard to use together. For example, a worker's skills might be written in a resume but not linked to a job post. As the platform grows, these sources need to be combined to make SerbiSure work smoothly.

Problem Identification

Existing integration problems include data silos where barangay records do not connect with Facebook hiring posts, making it hard to see the full picture. There is no standard format for resumes or job descriptions, leading to confusion. No central place exists for verifications, so checks are missed. Duplicate worker profiles happen when the same person signs up in different ways. Skills are spelled inconsistently, like "childcare" versus "alaga bata." Reports are

delayed because everything is put together by hand.

These issues make hiring risky and slow, as employers cannot trust the info, and workers miss opportunities. In a city like Cagayan de Oro, with rising demand for home help, this leads to unsafe matches and wasted time.

Why ETL is Necessary

An automated ETL pipeline is necessary to centralize scattered recruitment data, clean and make text standard, remove duplicates, check uploaded clearances, and create reports like employment ratios or trust scores. Without ETL, SerbiSure cannot give reliable matches, accurate community insights, or consistent verification records. This would keep the system as messy as current informal methods. Research shows that ETL helps platforms like this by improving data flow and trust.

Identify and Analyze Data Sources

Four data sources: JSON from worker registrations, JSON or text from job posts, Excel or CSV from barangay registries, and text from reviews.

Data Source	Format	Key Fields	Issues Observed
Worker Registration	JSON (Web Form)	Name, Address, Skills, NBI_file, Experience	Missing fields, inconsistent skill descriptions
Job Posts	JSON / Text	Employer_ID, Job_Description, Salary_Offer	Mixed English/Tagalog, vague descriptions
Barangay Registry	Excel (CSV)	Worker_Name, Barangay, Employment_Status	Duplicates, outdated records
Reviews	Text	Reviewer_ID, Rating, Comment	Bias, slang, sarcasm

Written Analysis

The potential data quality issues in these sources can harm SerbiSure. Missing values, like workers skipping experience fields, make profiles incomplete and hard to match. Inconsistent language, with mixed English and Tagalog, confuses the system when comparing skills or jobs. Duplicate records in barangay files or registrations lead to wrong counts of

available workers. Unstandardized salary formats, such as "5k" or "₱5000," prevent easy comparisons. Unstructured text in reviews, with bias or slang, makes it tough to measure trust fairly. These problems come from informal data entry and need fixing to make the platform reliable. Without handling them, matches could be wrong, and reports inaccurate.

Extraction Plan

Incremental Extraction will be used, pulling only new or updated records. This is chosen because SerbiSure is an ongoing platform with daily additions like new workers, reviews, and job posts. Pulling only changes improves speed and reduces strain.

Tools include Python with Pandas for CSV/Excel, No SQL needed, as sources are non-relational and target is Firebase Firestore; will use Firestore SDK for any queries if applicable., REST API calls for web data, and JSON parsing for forms.

Data will be validated during extraction by checking required fields like name and clearance upload, verifying file types for NBI, using timestamps for increments, and logging failures.

Why This Method is Appropriate

Incremental Extraction fits because data updates happen often but not all at once, keeping the system efficient for a live platform.

How Data Will Be Validated

We will scan for must-have fields, check document formats, and compare pull sizes to expected changes.

Potential Risks

Risks include API failures, handled by retry attempts; corrupt file uploads, fixed by validation checks; duplicate pulls, avoided with timestamps; and network interruptions, managed with logs and resume options.

Data Transformations

Transformations: Text normalization, duplicate removal, sentiment scoring.

Transformation Table

Transformation	What It Does	Why It Is Needed	Expected Output
Text Normalization	Standardizes skill and job text	Remove inconsistencies	Cleaned skill dataset
Duplicate Removal	Removes repeated worker records	Prevent double hiring records	Unique worker table
Sentiment Scoring	Converts reviews into trust score	Objective reputation system	Numeric trust rating

For Each Transformation:

1. Text Normalization (NLP Cleaning)

   Text normalization, an NLP cleaning process, solves the problem of mismatched descriptions where terms like “alaga bata” and “childcare” mean the same but are not recognized as such by the system. If not applied, the matching engine might fail to connect the right worker to the employer, leading to missed opportunities. This improves data quality by increasing matching accuracy and reducing mix-ups in meaning.

2. Duplicate Removal

   Duplicate removal addresses the issue of workers signing up multiple times, which creates extraneous records. Without this step, employment statistics would become inflated, causing confusion in hiring processes and decision-making. It enhances data quality by ensuring consistency and realism in the dataset.

3. Sentiment Score Generation

   Sentiment score generation tackles the subjectivity of review text, which is difficult to measure objectively. If omitted, trust scores would remain unclear and potentially unfair, affecting user reliability. This transformation boosts data quality by providing a fair, number-based rating system derived from the reviews.

Loading Strategy & Schema

Target Database

Firebase Firestore is chosen as the destination because it is a flexible, cloud-based NoSQL database that supports real-time syncing, handles semi-structured data like JSON uploads and text reviews efficiently, is scalable for web platforms, and provides built-in security rules for data integrity. It's ideal for SerbiSure's dynamic needs, like instant matching updates.

Written Explanation

Firebase Firestore is appropriate for SerbiSure as it stores data in collections of documents, making it easy to handle relationships between workers, jobs, and reviews through references or embedding. Data integrity will be maintained with security rules (e.g., requiring non-null fields), unique IDs via auto-generated keys, and validation in the ETL load process. We apply a denormalized structure to optimize reads, such as embedding frequent-access data like skills in worker documents, reducing joins for better performance in a web app.

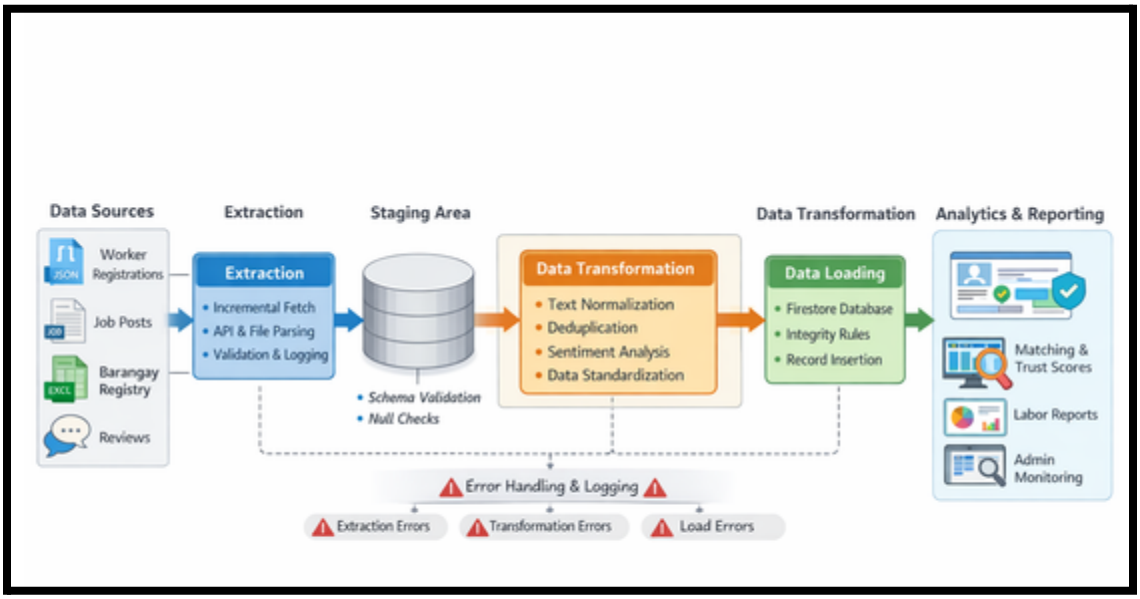
Schema Table

Field Name	Data Type	Description	Constraints
Worker_ID	String	Unique worker ID (auto-generated)	Primary Key (unique), Not Null
Full_Name	String	Worker name	Not Null
Barangay	String	Location	Not Null
Skills_Cleaned	Array (of Strings)	Standardized skills	Not Null
Trust_Score	Number	Sentiment score	Default 0
Clearance_Status	Boolean	Verified or not	Not Null

Explain

The schema design is appropriate because it leverages Firestore's document-based structure for flexible storage, enabling easy additions such as photo URLs or variable skills without rigid schema changes, while supporting quick queries essential for real-time matching in a web platform. Data integrity is maintained through Firestore security rules that enforce required fields and validations, unique auto-generated IDs that prevent duplicates, and app-level checks during the ETL load process to handle constraints effectively. The level of normalization used is denormalized, equivalent to first normal form (1NF), which embeds related data for improved read speed and avoids complex joins typical in NoSQL environments.

ETL Flow Diagram (Conceptual)



Literature Support

The design decisions made for the SerbiSure ETL pipeline are grounded in contemporary research on data quality management, ETL architecture, and sentiment analysis. This section synthesizes peer-reviewed studies published between 2019 and 2024 and directly connects their findings to SerbiSure's ETL design to demonstrate how research supports the system's data validation, transformation, and loading strategies.

Data Quality in ETL Processes

The challenges encountered in SerbiSure's fragmented data landscape—inconsistent skill descriptions, duplicate worker profiles, and missing fields—reflect widely documented problems in ETL-based systems. Souibgui et al. (2019) conducted a preliminary study on data quality within ETL processes and demonstrated that quality issues can be classified according to each stage of the pipeline: extraction, transformation, and loading. Their work shows that problems such as incomplete records at the source level, schema-level inconsistencies during transformation, and referential integrity violations during loading are pervasive across data integration projects. This multi-dimensional view of data quality directly informs SerbiSure's decision to implement validation checks at each ETL stage—requiring must-have fields such as the worker's name and clearance upload during extraction, applying text normalization and deduplication during transformation, and enforcing document-level validation rules during the load into Firebase Firestore.

Souibgui et al. (2019) further emphasized that ensuring data quality requires actively tracking quality defects along every phase of the ETL process, and that existing commercial ETL tools still fall short in consistently addressing all data quality dimensions. For SerbiSure, which relies on trustworthy worker profiles and verified clearances to enable safe hiring, this finding validates the platform's emphasis on proactive data integrity measures at every pipeline stage rather than correcting errors after the fact.

ETL Architecture and Data Warehousing Design

The choice of incremental extraction and a cloud-based NoSQL target database aligns with established principles in data integration design. Khan et al. (2024) emphasize that effective ETL architecture requires structured source identification, systematic transformation rules, and strict data quality controls when integrating heterogeneous formats such as databases, flat files, and APIs.

SerbiSure reflects this approach through timestamp-based incremental extraction, predefined transformations (text normalization, duplicate removal, and sentiment scoring), and loading into Firebase Firestore—a flexible, document-based NoSQL database organized into collections and documents. Data integrity is maintained through Firestore security rules that enforce required fields, auto-generated document IDs that prevent duplicate entries, and app-level validation checks applied during the ETL load process to ensure only clean, complete records are written to the database. This denormalized structure embeds frequently accessed data, such as skills, directly within worker documents, optimizing read performance for real-time matching in a web platform without the overhead of complex join operations.

Although some modern architectures adopt ELT (Extract, Load, Transform) models where transformation occurs after loading, SerbiSure deliberately implements a traditional ETL approach. Because the platform handles sensitive worker profiles and verification records, validating and transforming data before loading minimizes the risk of introducing incomplete or inconsistent records into the production database. This pre-load quality enforcement aligns with research highlighting the importance of data validation across all ETL stages (Souibgui et al., 2019) and supports the system's requirement for high-integrity, cloud-ready data before it reaches Firestore's live collections.

Sentiment Analysis for Reputation and Trust Scoring

SerbiSure's third transformation—converting raw review text into a numeric trust score—is grounded in research on sentiment analysis applied to online reviews. Vijayaragavan et al. (2024) demonstrated in a peer-reviewed study published in Scientific Reports that sentiment analysis of platform reviews can provide actionable, objective measures of user experience, particularly in service-oriented contexts where subjective language and informal expression are common. Their study introduced a Weighted Parallel Hybrid Deep Learning approach (WPHDL-SAEPR) applied to e-commerce product reviews and confirmed that automated sentiment scoring significantly improves the reliability and fairness of reputation metrics compared to raw star ratings or manual inspection.

This finding is directly applicable to SerbiSure's review pipeline, where comments from homeowners about domestic workers contain slang, sarcasm, and code-switching between Filipino and English. Without automated sentiment analysis, trust scores would remain vague and susceptible to bias. As Vijayaragavan et al. (2024) observed, transforming qualitative feedback into a quantifiable sentiment score enables platforms to offer consistent, data-driven reputation assessments—a goal central to SerbiSure's mission of creating safe and trustworthy hiring environments. These computed trust scores are stored as numeric fields within each worker's Firestore document, enabling real-time retrieval and display during the matching process without requiring additional query overhead.

Integration of Literature with Design Decisions

Taken together, these sources provide a rigorous scholarly foundation for the SerbiSure ETL pipeline. Souibgui et al. (2019) justify the multi-stage validation approach by showing that data quality defects occur at every phase of ETL and must be actively tracked—a principle directly reflected in SerbiSure's field-level checks during extraction, transformation rules during processing, and Firestore security rules during loading. Khan et al. (2024) support the architectural choices of incremental extraction, systematic transformation planning, and the use of a cloud-scalable, document-based storage system capable of handling heterogeneous source formats from JSON forms, Excel registries, and unstructured review text. Vijayaragavan et al. (2024) validate the sentiment scoring transformation as a method for producing objective, numeric trust metrics from subjective review text, directly informing how SerbiSure generates and stores reputation scores in Firestore worker documents. Each design decision in this paper can therefore be traced to credible, peer-reviewed evidence, ensuring that the SerbiSure ETL system is built on a sound theoretical and empirical basis.

Conclusion

The biggest data risks in the proposed SerbiSure system involve inaccurate, incomplete, and inconsistent data entering the pipeline. Since the platform collects information from web forms, barangay records, job posts, and text reviews, there is a high chance of missing fields, duplicate worker profiles, inconsistent skill descriptions, and unverified clearance uploads. If these risks are not controlled, the system may produce incorrect matches between homeowners and

domestic workers. There is also a risk that sentiment scoring may misinterpret mixed Filipino and English reviews, which could affect a worker's trust score. Because the platform is built to support safe hiring in Cagayan de Oro City, unreliable data directly threatens user trust and platform credibility.

If a transformation fails, the impact on the business and users can be serious. For example, if text normalization fails, skills such as "alaga bata" and "childcare" may not match, causing qualified workers to miss job opportunities. If duplicate removal fails, the system may count the same worker multiple times, leading to incorrect reports for barangay stakeholders. If sentiment scoring fails, trust scores may be unfair or inaccurate, which could damage a worker's reputation or mislead employers. In short, failed transformations reduce confidence in the system, slow down hiring decisions, and may result in unsafe matches.

An automated ETL process directly improves decision making because it provides clean, organized, and updated data in one central database. Instead of manually combining spreadsheets, chat messages, and paper records, the system automatically validates required fields, standardizes skills, removes duplicates, and calculates trust scores. This allows homeowners to quickly identify qualified and verified workers. It also allows barangay officials and the superadmin to view reliable reports on employment trends and verification status. With accurate and timely data, decisions are based on facts rather than guesswork.

The system can scale as data grows by continuing to use incremental extraction and a flexible cloud database such as Firebase Firestore. Incremental extraction ensures that only new or updated records are processed, which reduces system load as the number of users increases. Firestore supports automatic scaling, meaning it can handle more worker profiles, job posts, and reviews without requiring major redesign. Because the schema is flexible and semi structured, new fields such as additional verification documents or ratings can be added without disrupting existing data. This allows SerbiSure to grow alongside the increasing demand for domestic work services.

If this architecture were moved to the cloud using an ELT model instead of ETL, the main change would be the order of processing. In an ELT setup, raw data would first be loaded directly into the cloud storage system, and transformations such as normalization, duplicate removal, and sentiment scoring would occur afterward within the cloud environment. This would allow the system to store all original data before cleaning, which can be useful for advanced

analytics and auditing. Cloud platforms also provide stronger processing power, which can handle large volumes of text analysis more efficiently. However, this approach would require strict access controls to ensure that unprocessed sensitive data, such as clearance documents, remains protected. Overall, the system would become more flexible and powerful, but it would require stronger governance and monitoring to maintain data integrity and security.

References

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